Spatial simulations of tumor growth show driver mutations concentrate near boundary

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Abstract

Although a typical tumor will contain thousands of mutations, only a select few confer a selective advantage to cancer cells. These so-called "driver" mutations are responsible for uncontrolled growth and metastasis. Existing mathematical models of cancer progression usually assume that driver mutations are well-mixed throughout the tumor. Since cancer growth is inherently a three-dimensional process, these models are unable to capture spatial biases in the distribution of mutations. In this paper, we present a simple model of tumor growth and mutation that incorporates spatial constraints. Our algorithm avoids many of the pitfalls encountered by previous attempts to model 3-dimensional cell growth. As a result, we can efficiently generate and visualize tumors with millions of cells. Despite the simplicity, our model is able to address important questions about how spatial constraints impact the distribution of mutations. For instance, we find that the frequency of driver mutations near the tumor boundary is four times higher than the frequency near the center of the tumor. We briefly comment on the significance of our findings and discuss potential applications of the model.

Introduction

The cells within a cancerous tumor are usually highly diverse. An average tumor contains hundreds of thousands of mutations spread throughout billions of cancer cells [17]. Most of these mutations are harmless and do not contribute to the development of cancer. Only around 20 mutations are "drivers" and are responsible for facilitating the progression of cancer into later stages [28]. A lack of understanding about the evolutionary process which results in the observed intratumor heterogeneity is a major obstacle preventing the development of effective cancer therapies [27].

There is a rich history of mathematical models that attempt to quantify cancer progression. A common approach is to model cell growth as a stochastic branching process where cells randomly replicate and die at various rates [4]. To account for the heterogeneity found in populations of cancer cells, branching process models can easily incorporate the accumulation of mutations during cell division. Cells that acquire a driver mutation can be given a selective advantage by increasing the rate of division or decreasing the death rate. The fitness advantages are often drawn from a distribution or multiplied to the existing birth rate [3, 5, 12]. Branching process models usually assume that the population of cells is "well-mixed" in that mutants are evenly distributed throughout the tumor. The well-mixed assumption is important for obtaining models that are tractable

both analytically and computationally [4].

A major drawback of well-mixed models is that they ignore the fact that tumor growth is inherently a spatial process. The spatial distribution of (driver) mutations potentially impacts the dynamics of tumor growth. In addition, the distribution of mutants within the tumor may influence the rate in which metastases are seeded and can impact the results of clinical samples [18]. Many mathematical models have been developed to impose spatial constraints upon cancer growth [13, 15, 23]. The simplest models are based on a cellular automaton for cell growth [14] where cells occupy sites on the integer lattice and replicate into adjacent spaces. Since stochastic simulations of lattice based models are computationally intensive, a common decision is to simulate in two-dimensions instead of three-dimensions [8, 24]. However, the degree of mixing between clones is clearly dependent on dimension and thus a two-dimensional lattice model may underestimate intratumor heterogeneity.

The state of the art lattice-based model was introduced by Waclaw et. al. in [29]. Unlike previous models, the simulator of Waclaw et. al. places no restrictions on the number of mutations that can occur which makes it a useful tool for studying intratumor heterogeneity. The simulator is also claimed to generate 3D tumors of size 10^9 cells given 24 hours of run-time on a powerful desktop. However, this efficiency comes at the expense of using an approximate Gillespie algorithm where the cell division rate can be highly inflated. Moreover, the experiments of Waclaw et. al. are mainly focused on studying how driver mutations and local migration impact intratumor heterogeneity, leaving many questions about the spatial distribution of mutants unanswered.

In this paper, we develop a three-dimensional simulator of tumor growth and use it to study the spatial distribution of mutations. Our simulator improves on Waclaw et. al. in several ways. Algorithmically, we use an exact Gillespie algorithm which avoids inflated rates of cell division and allows for model parameters to directly correspond to well-mixed models. Through a careful implementation, we can simulate tumors of size 10^6 cells in 2 minutes which is on par with Waclaw et. al. Since usability is a high priority, we ensure that the entire simulation can be called and analyzed within R. We show that the growth rate of cells is polynomial in time and the location of the k-th mutant grows like an inverse cubic. We find that the concentration of mutants increases linearly as distance from the tumor origin increases. In particular, the tumor boundary is the most mutated region of the tumor. In the case of driver mutations, this implies that the most aggressively growing cells are located on the exterior of the tumor. This finding is potentially significant to the study of metastasis, as cells near the tumor boundary may be more likely to escape the primary tumor and seed secondary sites. We suggest that our model may also be useful for generating simulated sequencing data.

Methods

Model Formulation. We model tumor growth as an exponential birth-death process [2] where cells occupy sites on the three-dimensional integer lattice. Each cell is given a birth rate b and death rate d such that the time until cell replication or cell death is exponentially distributed with parameters b and d, respectively. A cell can replicate if at least one of the six sites adjacent to it is unoccupied. Each time a cell replication occurs, the daughter cell acquires a genetic alteration with some small probability u. We follow the "infinite alleles model" [20] in assuming that each alteration occurs only once, so that each mutation event induces a unique allele in the population.

An alteration is a driver mutation with probability u_d and is a passenger mutation otherwise. To model the selective advantage conferred to driver mutations, we assume that the birth rate of a cell depends on the number of driver mutations present in its genotype. Following a common assumption [4, 5, 29], a cell with k driver mutations is given birth rate bm^k where $m > 1$ and b is the initial, or wild-type, birth rate. The death rate d will remain the same among all cells, as the dynamics of the model are completely determined by b/d.

Since cancerous tumors are believed to often begin with a single mutated cell [9], the initial state of our

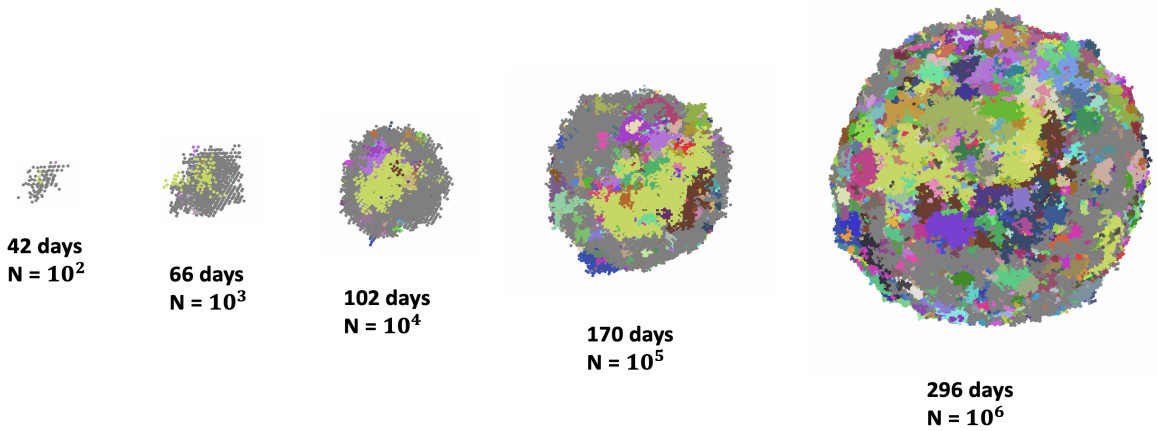

Figure 1: A demonstration of a tumor growing to size 10^6 cells for growth advantage $m = 1.1$. The wild-type allele is colored grey and all other unique alleles are assigned a random color.

model is a single cell at the origin with $b > d$.

Implementation. We simulate our model using a Gillespie algorithm [16]. Given a population of N cells at time t, cell i is chosen to replicate with probability $b_i / \sum_{j=1}^{N}(b_j + d_j)$ (provided a free space is available) and die with probability $d_i / \sum_{j=1}^{N}(b_j + d_j)$. After an event is selected, the time is updated to be $t + X$, where

$$X \sim \text{Exp}\left(\sum_{j \in V} b_j + \sum_{j=1}^{N} d_j\right) \tag{1}$$

and V is the subset of cells which are able to replicate (i.e, they have an unoccupied neighbor). Maintaining a list of the cells with a free neighbor can be overwhelming, so we approximate the parameter of (1) with $Np_{\max}$ where $p_{\max} = \max_j(b_j + d_j)$.

A standard approach is to use inverse transform sampling to select a cell for replication or death. However, this requires computing a cumulative sum over the birth and death rates for all unique alleles in the population, which is likely to scale linearly with N. This also requires that the data for each cell is organized by genotype. We circumvent these issues by using rejection sampling [7]. On each iteration, a random cell i is selected uniformly from the population. Next, we obtain a sample u from the uniform distribution over $[0, p_{\max}]$, where $p_{\max}$ is as above. If $u < b_i + d_i$, then cell i is selected to replicate or die. Otherwise, we proceed to the next iteration. Although there are contrived examples where rejection sampling can be inefficient, the expected run-time will be constant for reasonable parameter values.

In the interest of efficiency, the main simulation is written in C++. To make the code more accessible, the simulation can be called from an R script and the results will automatically be processed. Interactive visualizations of the simulated tumor can be obtained using the rgl package [1]. The software is still in development and will be made freely available sometime during summer 2020.

Results

An example visualizing the output of the simulator is given in figure 1. Although replication rates can vary widely between cancer types, [5] finds that a division rate of 1 cell every 4 days agrees reasonably well with clinical data. [5] also estimates passengers to occur at a rate of 0.016 mutations per division and drivers at a rate of 3.4×10^{-5} mutations per di-

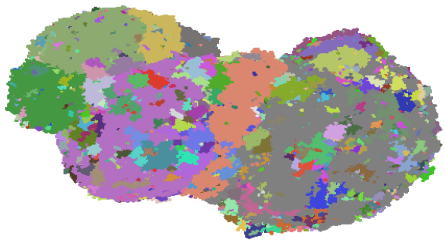

Figure 2: A tumor of size 5×10^5 for the large parameter value $m = 2$.

vision. To model aggressive tumor growth, [29] states that $b/d \approx 2$ leads to clinically detectable size in approximately six months. We therefore fix the model parameters at $b = 0.25$, $d = 0.13$, $u = 0.01$ and $u_d = 3 \times 10^{-5}/u = 3 \times 10^{-3}$. The parameter m controls the selective advantage conferred to driver mutations, and will vary between 1 (neutral growth) and 1.2 in the following experiments. For large values of m (i.e., $m \approx 2$), the direction of growth becomes biased and local heterogeneity appears to be drastically reduced, see figure 2.

Growth rate. In the well-mixed model, the expected number of cells at time t is known to be $e^{t(b-d)}$ in the absence of driver mutations [8]. In our model, the spatial constraints stifle exponential growth. By adapting techniques used to analyze contact processes [11, 22], it is possible to show that the expected number of cells at time t is $O(t^3)$. The main idea behind the proof is to upper bound the number of cells by the expected number of cells in the more tractable process where cell death never occurs. The bound generalizes to the analogous process on $\mathbb{Z}^n$, so that the expected number of cells at time t is $O(t^n)$.

Figure 3 (A) shows the growth curves for 100 trials and indicates that the population size is bounded above and below by a cubic function in t. This gives evidence that the $O(t^3)$ theoretical upper bound is tight. Interestingly, the presence of driver mutations does not appear to influence the asymptotic growth rate. Figure 3 (B) plots the log population size against log time for varying values of m. As m increases, the curves remain linear and of the same slope, indicating that growth is polynomial of the same degree. The increasing birth rate due to driver mutations certainly impacts the constant associated with cubic growth, although the precise dependence of this constant on the parameters is currently unknown.

Spatial distribution of mutants. We are particularly interested in studying the spatial biases in the distribution of mutations that emerge from our model. A rough estimate yields that the k-th mutation occurs at a radius of $O(k^{1/3})$ from the origin. To see this, note that the distance from the origin of the k-th mutation is bounded by the radius of the tumor at the time in which the mutation occurs. Since growth is approximately spherical, the growth rate bound from the previous section gives us that the radius of the tumor grows linearly in t. Since mutations occur on average once every $1/u$ births, the time of the k-th mutation should grow inverse cubically. In the case of neutral growth, figure 4 (A) confirms the estimated bound.

Next, we analyze how the number of mutations per cell changes as we move away from the origin. Figure 4 (B) shows that the average number of mutants per cell grows linearly as the radius from the origin increases and that the rate of growth does not appear

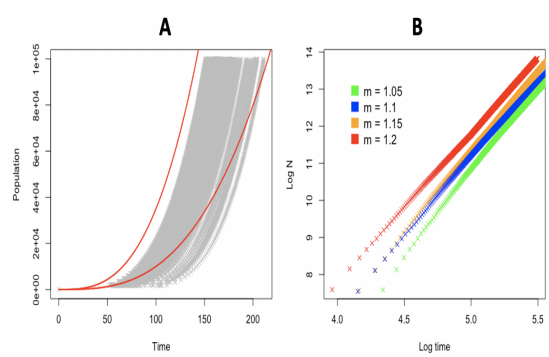

*Figure 3: **(A)** In grey, the growth curves for 100 simulations to a population size of 10^5 cells. In red, the plots of $y = 0.035t^3$ and $y = 0.01t^3$. **(B)** $\log N$ vs $\log t$ for simulations to 10^6 cells using varying m.*

to depend strongly on m. Restricting our attention to driver mutations only, figure 4 (C) shows the number of driver mutations per cell can spike near the tumor boundary when m is large.

We also investigate how distance impacts genetic diversity. Given the genotypes g_1 and g_2 of two cells,[1] the *Jaccard similarity* between the two cells is defined to be $|g_1 \cap g_2|/|g_1 \cup g_2|$. Figure 4 (D) shows that the average Jaccard similarity decreases linearly ($p < 0.01$) as the Euclidean distance between the two cells increases. The presence of driver mutations does not appear to alter these observations. We also measure the heterogeneity in different spherical "shells" within the tumor. Given two cells at positions ϕ_1 and ϕ_2 (in $\mathbb{R}^3$) with $||\phi_1|| \approx ||\phi_2||$, we define the *normalized spherical distance* between the two cells to be the arc length between $\phi_1/||\phi_1||$ and $\phi_2/||\phi_2||$ on the unit sphere. Note that this can be computed as $2 \arcsin(\Delta/2)$, where Δ is the Euclidean distance between $\phi_1/||\phi_1||$ and $\phi_2/||\phi_2||$. Figure 4 (E) plots the mean Jaccard similarity against the normalized spherical distance for various shell radii. As the radius increases, the genetic diversity between cells in the same spherical shell increases as well.

A physical definition of the tumor boundary is that they are the cells which can be "reached" from outside the tumor. To make this precise, let $p^* \in \mathbb{Z}^3$ be any site outside the convex hull of the tumor. Then a cell $c \in \mathbb{Z}^3$ is on the boundary of the tumor if there exists a path from c to p^* traveling along vacant adjacent sites. We compute the set of boundary cells in linear time by viewing the lattice as a graph and applying a breadth first search [10]. For comparison,

[1] For our purposes a genotype is the set of mutations present in the cell

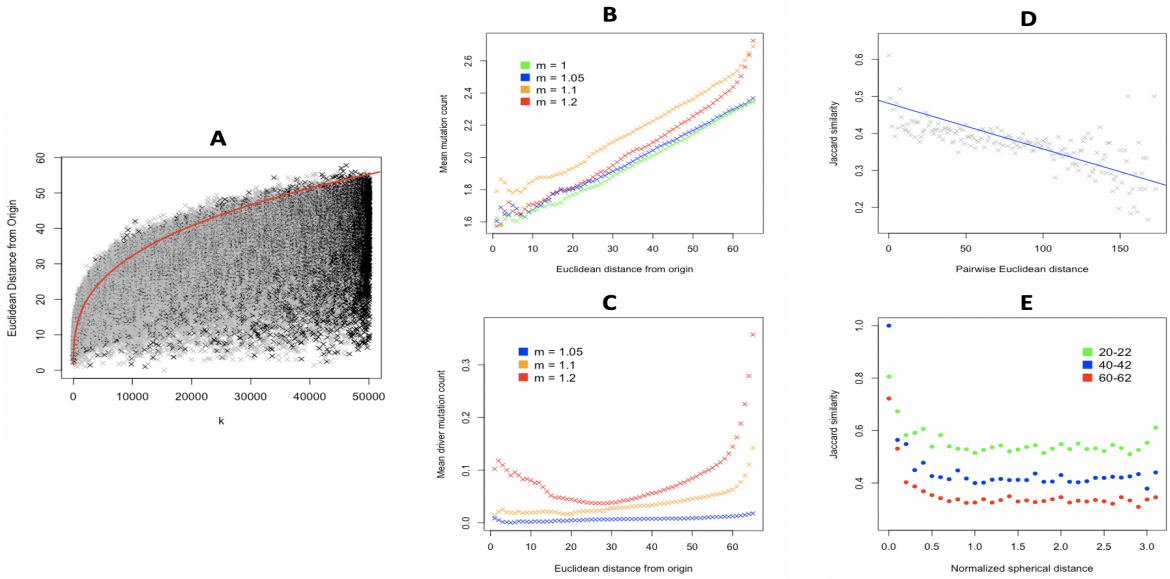

Figure 4: Quantifying the spatial distribution of mutants. ***(A)*** *The radial distance of the k-th mutant, tumor size* 10^6*. Point is grey if the mutant becomes extinct before simulation is stopped and is black otherwise. In red, inverse cubic fit.* ***(B)*** *Mean number of mutations per cell plotted against Euclidean distance from origin. Mean of* 10 *trials for tumor size* 10^6*.* ***(C)*** *Same as previous plot, except for driver mutations only.* ***(D)*** *Mean Jaccard similarity plotted against Euclidean distance for* $15,000$ *randomly sampled pairs of cells.* $m = 1.1$ *and* $N = 10^6$*.* ***(E)*** *Mean Jaccard similarity plotted against normalized spherical distance for spherical shells of varying radii.* 10000 *randomly sampled pairs of points for each shell size and* $m = 1.1$*,* $N = 10^6$*.*

m	**Interior**	**Near boundary**	**Boundary**
1.01	0.0016	0.0044	0.0047
1.05	0.0029	0.0087	0.0082
1.1	0.0134	0.0495	0.0404
1.2	0.0755	0.4245	0.2090

Table 1: Frequency of driver mutations in different sections of the tumor. Mean of 10 *runs with* $N = 10^6$.

we also define the interior of the tumor to be the set of cells that have radius less than $0.5r_{\max}$, where $r_{\max}$ is the largest radius among all cells, and a section of cells near the boundary which consists of cells with radius greater than $0.75r_{\max}$. Table 1 shows the frequency of driver mutations in each of the three cell categories for various values of m.

Discussion

Our model shows that heterogeneity increases as radial distance to the origin increases. Consequently, cells near the tumor boundary have acquired a diverse range of mutations, many of which confer selective advantage. Figure 4 (C) indicates that the number of driver mutations per cell can spike at large distances from the center, which suggests that the exterior of the tumor may consist of the most aggressively growing cells. Table 1 confirms this by showing that the frequency of driver mutations near the tumor boundary is nearly four times higher than the frequency of drivers in the interior of the tumor.

This observation may have important implications for cancer metastasis. While the primary tumor grows, some of the cancer cells are shed into the bloodstream. The cells that leave the primary tumor may travel to distant sites and seed secondary tumors [25]. Aggressive tumors can shed thousands of cells per day, although only a select few survive long enough to seed a metastasis [6]. While the cells that can be shed depends on the tumor vasculature (which we do not model), it might be reasonable to assume that cells near the boundary are the most likely to escape the primary tumor and intravasate into neighboring blood vessels. This would suggest that cells released into circulation are highly heterogeneous, an observation in line with experimental evidence [26]. Moreover, the escaping boundary cells have a higher frequency of driver mutations which could mean that the most aggressively growing cells are the ones seeding secondary sites. An interesting future direction is to combine spatially biased seeding with existing methods for inferring the diversity of metastases such as [18]. Additionally, we could model the role of blood vessels by having lattice sites convey fitness to cells based on their proximity to vessels and to have cell division be biased in the direction of vessels.

A potential application of our model is the simulation of sequencing data. Understanding the evolutionary history of a given tumor is an important problem that could be useful for targeted therapy [31]. The recent advancement of single cell sequencing, which allows us to observe the complete genotype of a specific cell, may hold the key to solving this problem. A common approach is to use the single cell data to infer a phylogenetic tree for the cells within the tumor [30]. Inferring an accurate tree from data is a hard problem since the chosen algorithm must be resilient to large amounts of noise due to sequencing technology [30]. Recently, there have been many methods proposed to overcome this issue (e.g., [19] and [21]). However, little attention has been given to the fact that these algorithms might not be robust to spatially biased sampling. For this reason, simulated single cell data from our model could provide a benchmark to assess the quality of the various methods, and may provide insights into improvements that could be made. Additionally, the number of sampled cells required to obtain an accurate snapshot of the tumor history is not well known [30]. Sequencing more cells comes at an increased cost, but will ensure that cells from smaller clones are selected. We believe that simulated data from our model will be well suited for answering this question.

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